



POLSCI 9592

Lecture 1: Maximum Likelihood Estimation

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Goals for this class

1. Introductions
2. Discuss direction for the course.
3. Describe Maximum Likelihood Estimation (MLE)
4. Consider a couple of examples of MLE

Example: London Mayoral Election

London is having a mayoral election. Incumbent (and former Western LG Program Administrator) is running.

- A poll shows:

	Voters
Other	48
Morgan	52
Total	100

Assess H_A : Morgan wins more than 50% of the vote.



How to Figure This Out?

- **What are we trying to find?** The population parameter π^M — the probability a voter picks Morgan.
- **What do we need to learn about it?** A point estimate, and some sense of how much that estimate could vary from sample to sample.
- **Where does the estimate come from?** The natural best guess for π^M :

$$\text{Proportion for Morgan} = \frac{\text{Number for Morgan}}{\text{Total Respondents}}$$

$$p^M = \frac{N^M}{N^T}$$

Uncertainty

- **Where does the uncertainty/sampling distribution come from?** Rather than deriving it directly, we get it for free once we've modeled the dependent variable itself.
 - Think of each individual vote y_i (Morgan/Other) as the dependent variable. What do we know about it?
 - It only takes two values $\rightarrow$ binary/Bernoulli.
 - Each voter has the same underlying probability π^M — we have no information to say otherwise.
 - Secret ballot $\rightarrow$ one vote doesn't affect another $\rightarrow$ independence.
 - Those three properties mean $y_i \sim \text{Bernoulli}(\pi^M)$.
 - The total count $N^M = \sum y_i$ is a sum of iid Bernoulli trials, so $N^M \sim \text{Binomial}(N^T, \pi^M)$.
 - Since p^M is just N^M / N^T , this binomial distribution is (up to rescaling) the sampling distribution of our estimate.

Binomial Distribution

$$N^M \sim \text{Binomial}(N^T, \pi^M)$$

The p -value we could calculate using the Binomial PMF:

$$f(y) = \binom{n}{y} p^y (1 - p)^{(n-y)}$$

where $\binom{n}{y}$ is the binomial coefficient and is defined as $\frac{n!}{y!(n-y)!}$. Calculating this out tells us the number of possible outcomes of size n that have exactly y successes.

$$\Pr(N^M \geq 52) =$$

$$\sum_{k=52}^{100} \binom{100}{k} .5^k (1 - .5)^{(100-k)}$$

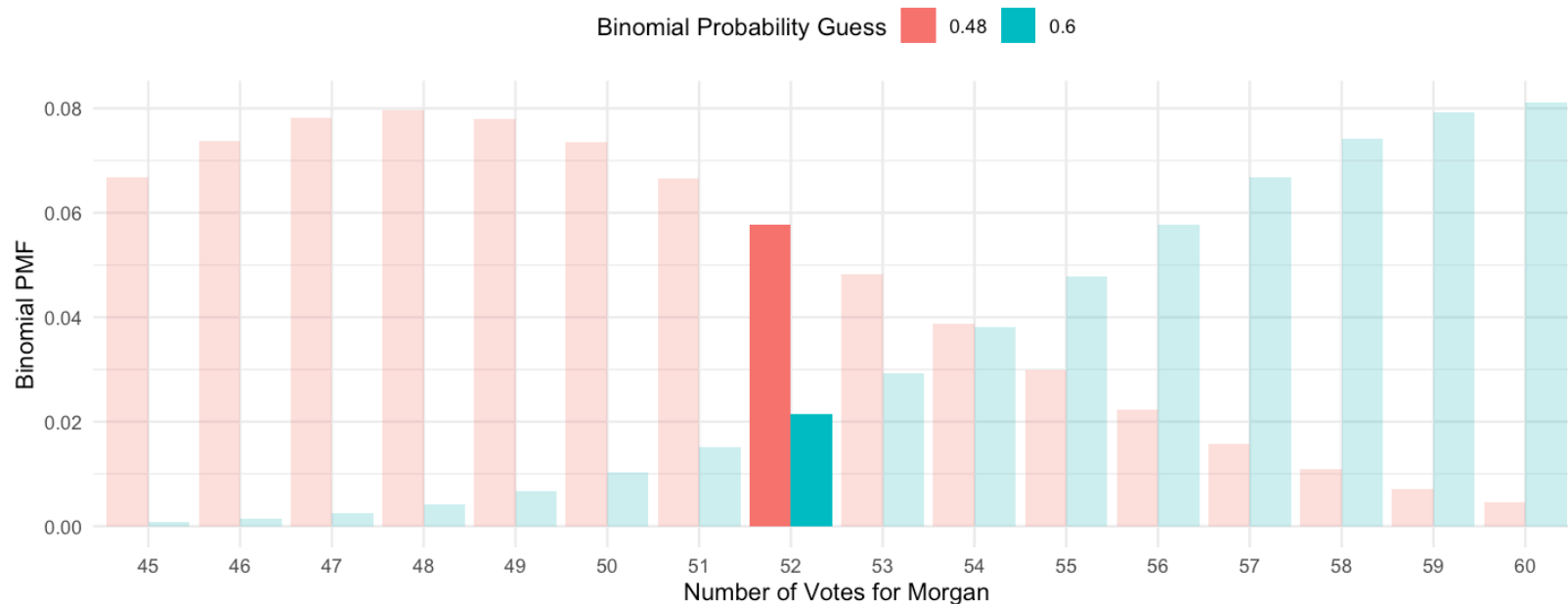
```
pbinom(51, 100, .5, lower.tail=FA
```

```
## [1] 0.3821767
```

How Did We Make a Best Guess?

Intuitively - sample proportion is best guess of population proportion.

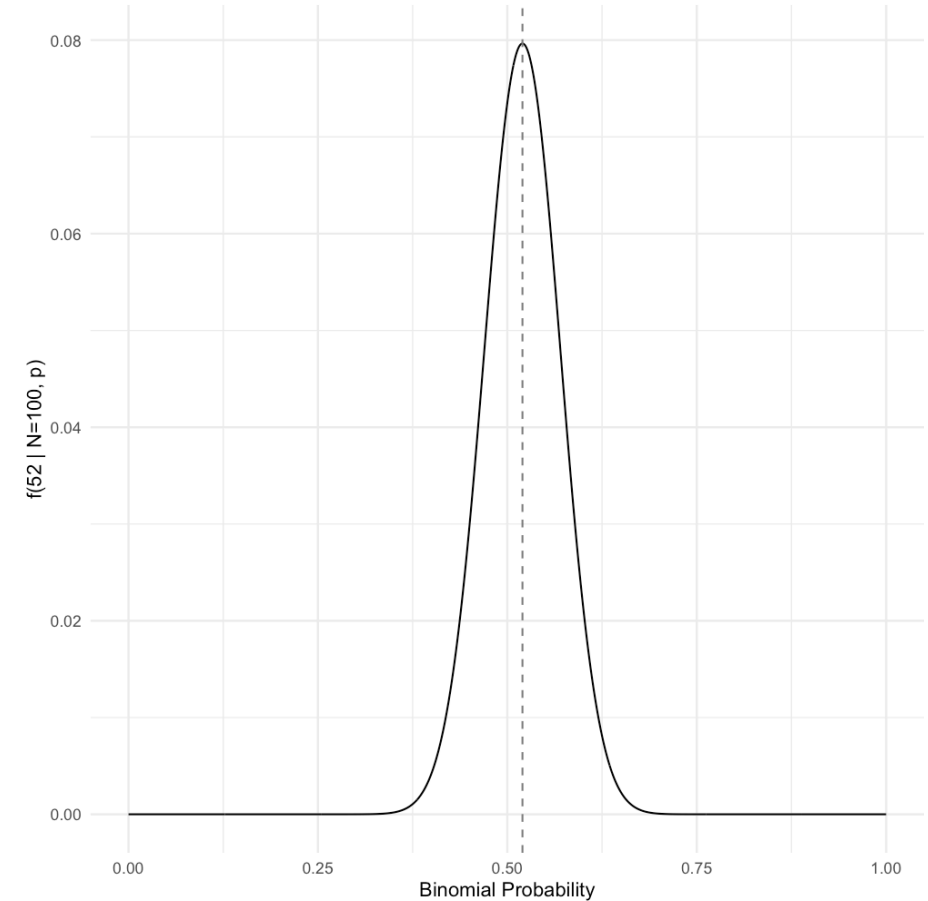
Systematically - First, if you had to choose between either .48 or .6 which is better?



All Values

Now, what if we looked at all possible values of the binomial probability?

- Which one maximizes the probability we would observe 52?

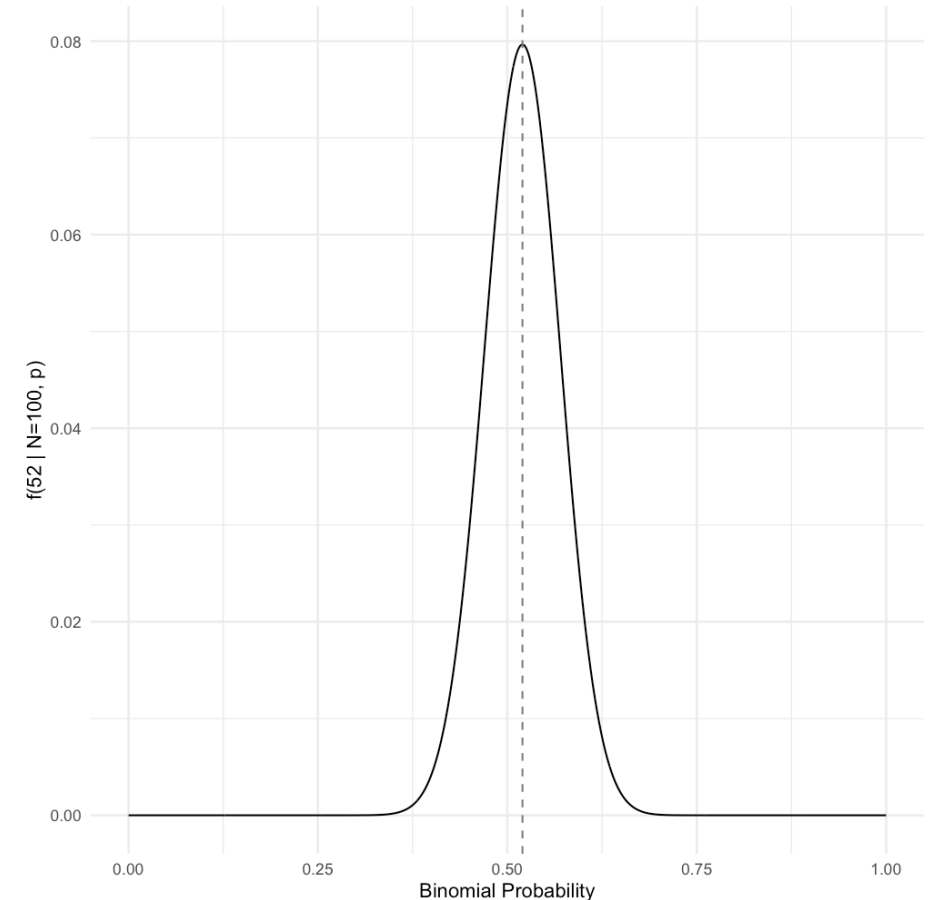


All Values

Now, what if we looked at all possible values of the binomial probability?

- Which one maximizes the probability we would observe 52?

We just did maximum likelihood estimation!



Maximum Likelihood Estimation (MLE)

Maximum Likelihood Estimation is a different way of estimating statistical relationships.

- Here, we try to identify which parameter values make it most likely (i.e., *maximize the likelihood*) that we would observe the data that we observe.

Probability

Before we go into MLE, we need to refresh ourselves on the axioms of probability. Let's assume that the sample space is S (the set of all possible outcomes):

1. For any event A , $Pr(A) \geq 0$.
2. $Pr(S) = 1$
3. If events A , B and C are mutually exclusive, then
$$Pr(A \cup B \cup C) = Pr(A) + Pr(B) + Pr(C)$$

Each statistical distribution has associated probability functions - most useful for right now - the PDF/PMF and the CDF.



Probability Density Function

One of the main elements of a likelihood estimation is a *probability density function* or *PDF*.

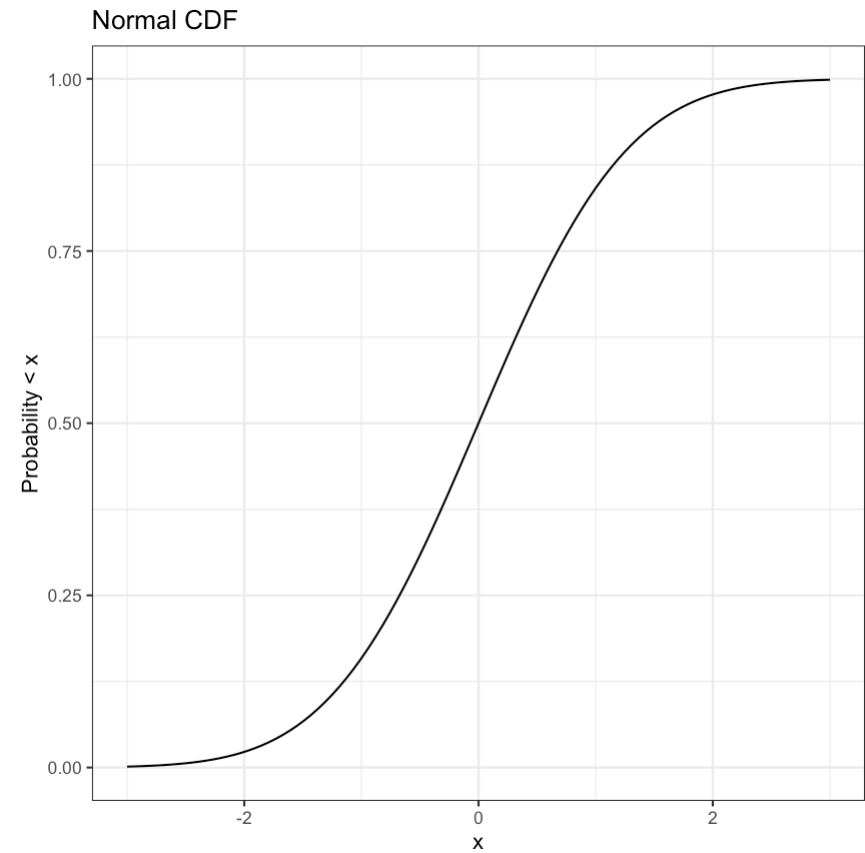
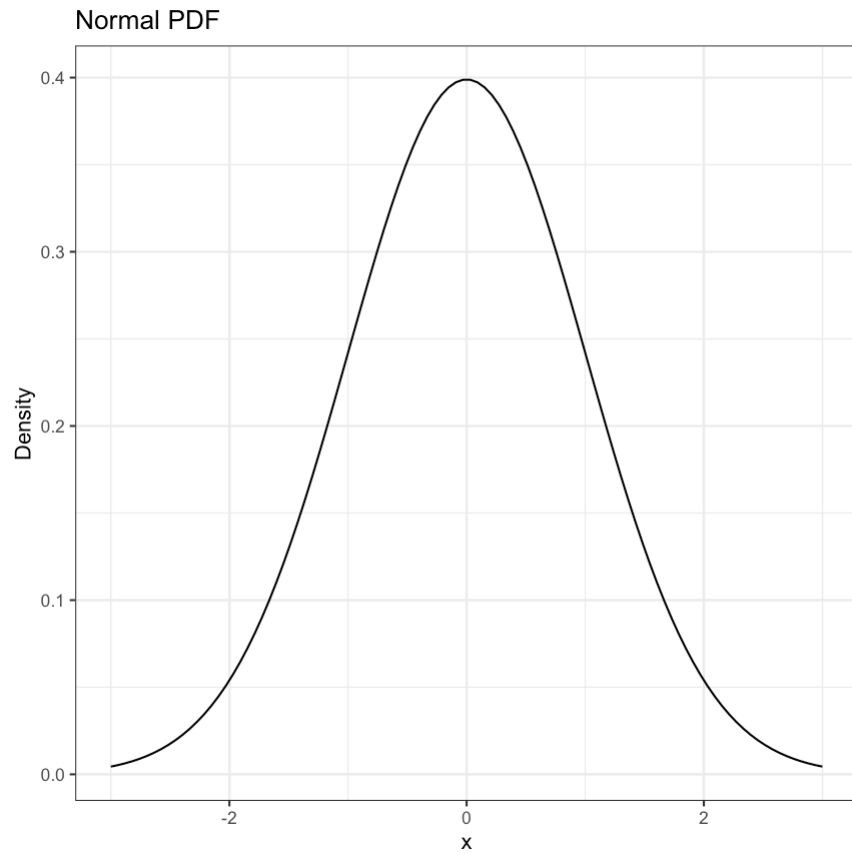
- The PDF gives the *relative likelihood* that an observation drawn randomly from a continuous random variable would take on a particular value.
- We can use these relative likelihoods to find the best parameters for our relationships.
- For discrete variables (like our binomial example), the distribution is a *probability mass function* or *PMF*.

Cumulative Distribution Function

A counterpart to the PDF is the *Cumulative Distribution Function* or *CDF*.

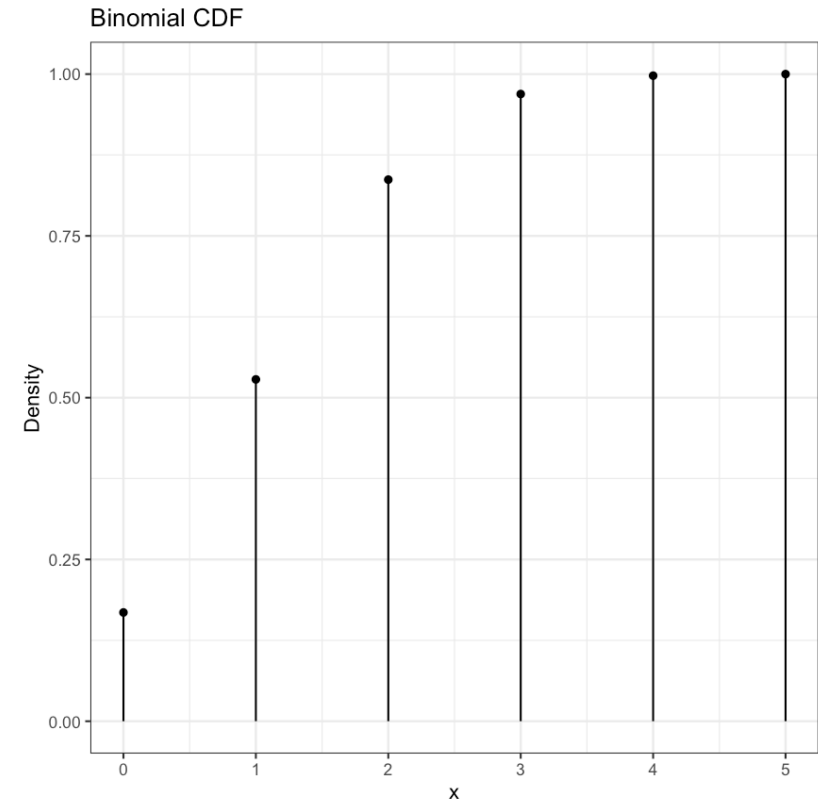
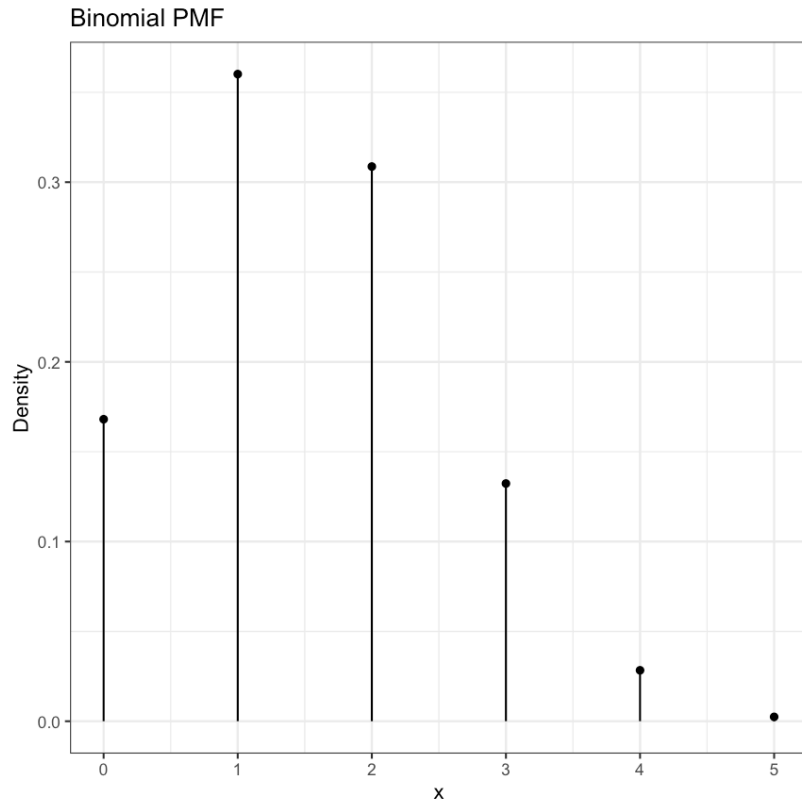
- The CDF tells us the probability of being below (or alternatively above) a certain value of a random variable.
- The z - and t - tables in the back of intro stats books give you the CDF for the normal and t distributions evaluated at lots of different points.

Normal PDF and CDF



Discrete Distributions

With discrete distributions (those where some values are impossible, like counts), we have a *probability mass function* or *PMF* instead of a PDF.



What is Likelihood

The Likelihood Axiom is as follows:

$$L(\tilde{\theta}|y) \propto f(y|\tilde{\theta})$$

- $f(y|\tilde{\theta})$ is a probability density function of y given the hypothetical model parameters $\tilde{\theta}$
- $\propto$ here means we could multiply $f(y|\tilde{\theta})$ by *any* positive function of the data alone and get the same answer for which $\tilde{\theta}$ maximizes it. The absolute value of L is never meaningful, only how it moves as $\tilde{\theta}$ changes.
- What Maximum Likelihood Estimation does is to pick the parameters $\hat{\theta}$ that make the data most likely to have been generated given the assumptions we make.

Evaluating the Likelihood function

Assuming that you've got lots of y values (i.e., y_i for $i = 1, \dots, n$), then you would want to know the aggregate likelihood for all values (i.e., a single number).

$$L(\theta|y) = \prod_{i=1}^n L(\theta, y_i) \propto \prod_{i=1}^n f(y_i|\theta)$$

While this would theoretically work fine, taking the product of a bunch of small numbers is going to generate something that the computer will have difficulty dealing with, thus we usually try to maximize the log-likelihood (ℓ).

$$\ell(\theta|y) = \sum_{i=1}^n \ell(\theta, y_i) \propto \sum_{i=1}^n \log(f(y_i|\theta))$$

Score Function

The score function $S(\theta)$ is the first partial derivative of the log-likelihood with respect to the parameter

- First derivative shows us how steep the log-likelihood is at any particular point.
- Setting the first derivative to zero gives us the point at which the likelihood is maximized.

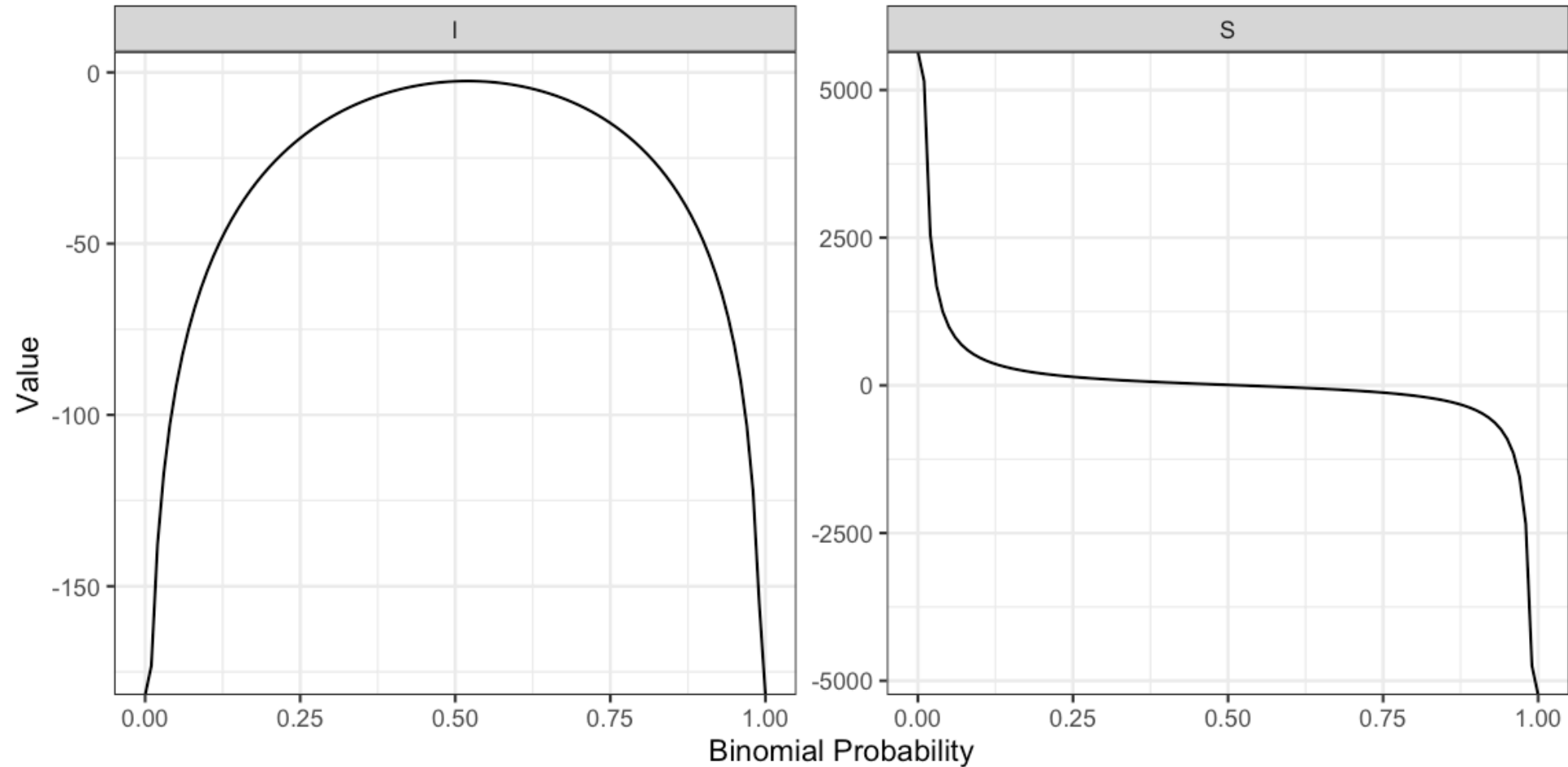
For the binomial model:

$$\ell(p) = k \log(p) + (n - k) \log(1 - p)$$

$$S(p) = \frac{\partial \ell}{\partial p} = \frac{k}{p} - \frac{n - k}{1 - p}$$



Likelihood and Score Functions





Math

First term — $\frac{d}{dp} [52 \log p]: \frac{d}{dp} [52 \log p] = 52 \cdot \frac{1}{p} = \frac{52}{p}$

Second term — $\frac{d}{dp} [48 \log(1 - p)]:$ here the thing inside the log isn't p itself but

$u = 1 - p$, so you need the chain rule: $\frac{d}{dp} \log(u) = \frac{1}{u} \cdot \frac{du}{dp}$. Since $\frac{du}{dp} = \frac{d}{dp} (1 - p) = -1$,

$$\frac{d}{dp} [48 \log(1 - p)] = 48 \cdot \frac{1}{1 - p} \cdot (-1) = -\frac{48}{1 - p}$$

Add them (sum rule):

$$S(p) = \frac{52}{p} - \frac{48}{1 - p}$$



Solve

$$\frac{k}{p} - \frac{n-k}{1-p} = 0$$

$$\frac{k}{p} = \frac{n-k}{1-p}$$

$$k(1-p) = (n-k)p$$

$$k - pk = pn - pk$$

$$k = pn$$

$$\frac{k}{n} = p$$

Sampling Uncertainty

Fisher Information is what tells us about the variability of the parameter estimate in repeated sampling.

- Negative expected first derivative of the score function with respect to p (negative expected second derivative of the log-likelihood function).
- A more peaked log-likelihood function has lower sampling variability.

Calculating Fisher Information I

$$S(p) = \frac{k}{p} - \frac{n-k}{1-p}$$

Fisher information is $I(p) = -E \left[\frac{d^2 \ell}{dp^2} \right]$, so first differentiate the score again:

$$\frac{d^2 \ell}{dp^2} = \frac{d}{dp} \left[\frac{k}{p} - \frac{n-k}{1-p} \right] = -\frac{k}{p^2} - \frac{n-k}{(1-p)^2}$$

Negate and take the expectation, using $E[k] = np$ since $k \sim \text{Binomial}(n, p)$:

$$I(p) = E \left[\frac{k}{p^2} + \frac{n-k}{(1-p)^2} \right] = \frac{np}{p^2} + \frac{n-np}{(1-p)^2} = \frac{n}{p} + \frac{n(1-p)}{(1-p)^2} = \frac{n}{p} + \frac{n}{1-p}$$

Calculating Fisher Information II

Combine over a common denominator:

$$I(p) = n \left[\frac{1}{p} + \frac{1}{1-p} \right] = n \cdot \frac{(1-p) + p}{p(1-p)} = \frac{n}{p(1-p)}$$

The inverse of Fisher Information ($I(p)^{-1}$) is the sampling variance of p

$$\text{Sampling Var}(p) = \frac{p(1-p)}{n}$$



I thought this was an applied class!

I showed you more math than usual here because it is "easy" (tractable) in this case.

Generally, this sort of closed form solution doesn't work out.



MLE in R

```
library(maxLik)
llfun <- function(par, x, n=100){
  p <- dbinom(x, n, par[1], log=TRUE)
  sum(p)
}
out <- maxLik(llfun, start=.5, x=52)
summary(out)
```

```
## -----
## Maximum Likelihood estimation
## Newton-Raphson maximisation, 1 iterations
## Return code 1: gradient close to zero (gradtol)
## Log-Likelihood: -2.530081
## 1 free parameters
## Estimates:
##      Estimate Std. error t value Pr(> t)
## [1,]  0.52000    0.04996   10.41  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## -----
```

```
# k/n
52/100
```

```
## [1] 0.52
```

```
# sqrt([p(1-p)]/n)
sqrt((.52*(1-.52))/100)
```

```
## [1] 0.04995998
```



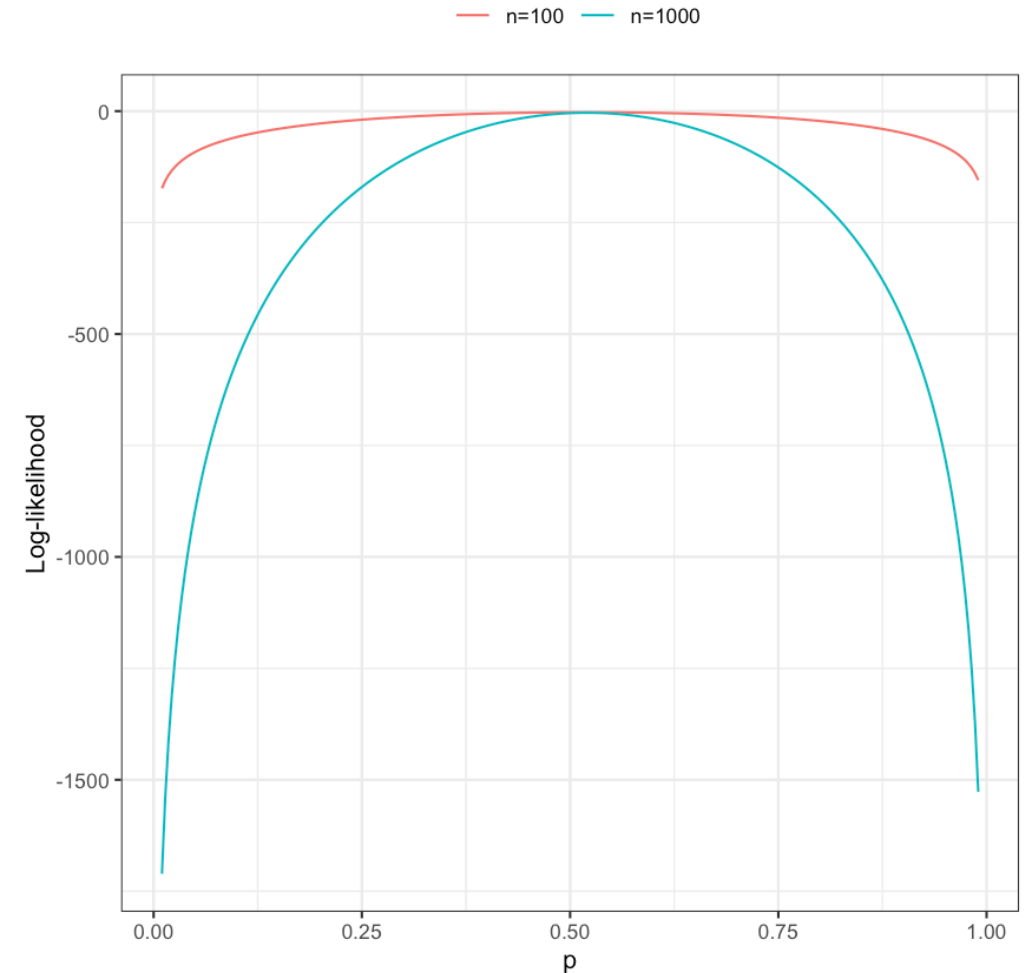
What if $N = 1000$?

```
out <- maxLik(llfun, start=.5, x=520, n=1000)
summary(out)
```

```
## -----
## Maximum Likelihood estimation
## Newton-Raphson maximisation, 2 iterations
## Return code 1: gradient close to zero (gradtol)
## Log-Likelihood: -3.679119
## 1 free parameters
## Estimates:
##      Estimate Std. error t value Pr(> t)
## [1,]   0.5200    0.0158   32.91 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## -----
```

Log-Likelihood Functions

We could look at the two likelihood functions, one from 100 points and one from 1000 points.





Owners vs. Renters

What if Homeowners and Renters have different opinions

	Renter	Owner	Total
Other	10	38	48
Morgan	20	32	52
Total	30	70	100

```
llfun2 <- function(par, x, n){
  l <- dbinom(x, n, par, log=TRUE)
  sum(l)
}
out2 <- maxLik(llfun2,
               start=c(renter=.5, owner=.5),
               x=c(20, 32), n=c(30, 70))

summary(out2)

## -----
## Maximum Likelihood estimation
## Newton-Raphson maximisation, 2 iterations
## Return code 1: gradient close to zero (gradtol)
## Log-Likelihood: -4.227176
## 2 free parameters
## Estimates:
##      Estimate Std. error t value Pr(> t)
## renter  0.66667   0.08607   7.746 9.48e-15 ***
## owner   0.45714   0.05954   7.678 1.62e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## -----
```



Testing the Two Models: Likelihood Ratio Test

If two models are nested, then we can use a likelihood ratio test to figure out whether the bigger one is "better".

$$s = -2 (\ell(M_{\text{small}}) - \ell(M_{\text{big}}))$$

Under H_0 : Both Models Same,

$$s \sim \chi^2_{k_{\text{big}} - k_{\text{small}}}.$$

```
llfun1 <- function(par, x, n){  
  l <- dbinom(x, n, c(par[1],par[1]), log=TRUE)  
  sum(l)  
}  
out1 <- maxLik(llfun1, x=c(20,32), start=c(both = .5), n = c(30,  
lmtest::lrtest(out1, out2)
```

```
## Likelihood ratio test  
##  
## Model 1: out1  
## Model 2: out2  
##   #Df  LogLik Df  Chisq Pr(>Chisq)  
## 1    1 -6.1036  
## 2    2 -4.2272  1  3.7529   0.05272 .  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Properties of MLEs

- Consistent - As sample size increases the probability that the MLE differs from the true parameter by an arbitrarily small amount is zero
- Asymptotically efficient which means that the MLE's variance is the smallest among all possible consistent estimators.
- Asymptotically normally distributed

All of these are asymptotic properties, so they describe the properties of the estimators as n is close to ∞ . They may behave differently in small samples. We will discuss this a bit later on.

Recap

1. What is MLE?

- $L(\theta|\text{Data}) \propto f(\text{Data}|\theta)$